# AAVAIL Capstone — Peer Review Solution

Practical checklist and implementation plan for all criteria shown in the peer-review submission.

## Submission Fields

| Field | Recommended entry |
|---|---|
| Project Title | AAVAIL Customer Churn Prediction & Model Deployment |
| GitHub URL | PASTE YOUR PUBLIC GITHUB REPOSITORY URL HERE |

## Peer Review Criteria

| # | Requirement | What to implement/show |
|---|---|---|
| 1 | 1. API unit tests | Use pytest to test health/status, single-country prediction, all-country prediction, status codes, and response structure. |
| 2 | 2. Model unit tests | Test preprocessing, feature creation, model loading, and prediction. Include valid and invalid inputs. |
| 3 | 3. Logging unit tests | Verify that a request creates the expected structured log fields: request/input summary, model version, prediction, times |
| 4 | 4. Single test command | Use `pytest -q` or `make test`. All API, model and logging tests should run together and pass. |
| 5 | 5. Performance monitoring | Log model version, timestamp, runtime, input summary and predictions. Provide a monitoring script/notebook to summa |
| 6 | 6. Isolate tests from production | Use pytest fixtures, temporary directories and mock/fixture models. Test logs must not overwrite production logs. |
| 7 | 7. API behavior | Support both a specific country, e.g. `/predict?country=US`, and combined/all-country prediction, e.g. `/predict`. Docume |
| 8 | 8. Automated ingestion | Implement a reusable `ingest_data()` function or `scripts/ingest.py` that loads, validates, combines and cleans the sourc |
| 9 | 9. Multiple models | Compare a baseline with multiple classifiers using the same split and metrics such as precision, recall, F1 and accuracy |
| 10 | 10. EDA visualizations | Include missingness, churn/class distribution, country comparison and predictor-vs-churn plots with clear titles and axes |
| 11 | 11. Dockerization | Include Dockerfile and requirements.txt. Build and run the API from the image and document the commands. |
| 12 | 12. Model vs baseline visualization | Create a bar chart or table comparing baseline and selected model metrics, especially F1/precision/recall for classificati |

## Recommended GitHub Structure

```
aavail-capstone/
app/
api.py
model.py
preprocessing.py
logging_utils.py
scripts/
ingest.py
train.py
monitor.py
tests/
test_api.py
test_model.py
test_logging.py
conftest.py
notebooks/
EDA.ipynb
model_comparison.ipynb
Dockerfile
requirements.txt
pytest.ini
README.md
Makefile
```

## Recommended commands

```
pytest -q
docker build -t aavail-capstone .
docker run -p 8000:8000 aavail-capstone
```

## README Evidence

Document the API examples, ingestion command, test command, Docker build/run commands, model comparison metrics, EDA plots, monitoring approach, and a short explanation of how test fixtures prevent writes to production models/logs.

## Important

This PDF is a submission-ready implementation checklist; it does not claim that your current GitHub repository already satisfies these items. Verify each feature in the actual repository before submitting the peer review.